Insertion-Deletion & Predicate Reads (Phantoms) DBMS

> **Definition:** The **Phantom Phenomenon** occurs when a concurrent transaction inserts or deletes tuples satisfying a search predicate while another transaction is executing range queries over the same predicate.

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1. Detailed Technical Explanation

1. The Phantom Problem Scenario

Suppose Transaction **T_1** reads all employees in `Dept_ID = 10` twice within its execution:

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Standard tuple-level locking fails to prevent this because the new tuple `106` did not exist when **T_1** acquired its locks!

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## 2. Solutions to Phantom Phenomenon

### 1. Predicate Locking (Logical Locking)

- Transaction **T_1** acquires a lock on a **predicate condition** (e.g., `Dept_ID = 10`).

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### 2. Index Locking (Physical Granularity Solution)

- If an index exists on `Dept_ID`, **T_1** locks the **index bucket / page** corresponding to `Dept_ID = 10`.

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### 3. Next-Key Locking (B+ Tree Index Lock)

- Locks both the target index record AND the gap immediately preceding/following it in the B+ Tree index, preventing insertions into the range.

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## 3. Core Concepts & Memory Keywords

- **Phantom Tuple:** A newly inserted row appearing in a repeated range query.

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4. Must-Write Points for Exams

- Standard tuple locks cannot prevent phantom rows because phantom tuples do not exist at lock request time.

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5. Quick Recall Flow

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Range
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